



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	16 FEBRUARY 2022
GRADE	11	MARKS	50
EXAMINER NAME	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
CLASS		DURATION	1 HOUR

INSTRUCTIONS:

1. This paper consists of 7 pages. Please check that your question paper is complete.
 2. Answer all questions on the folio paper provided.
 3. Write in blue or black pen only.
 4. Read all questions carefully and answer in as much detail as possible.
 5. All diagrams should have a clear heading.
 6. It is in your own interest to write legibly and to present your work neatly.
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QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	10	
2	17	
3	12	
4	11	
TOTAL: 50		

Question 1:

- 1.1. Various possibilities are suggested as answers to the following questions. Indicate the correct answer by writing the correct letter from the corresponding question on the folio paper provided. (5)

1.1.1. The kidneys filter and remove...

- A. urea
- B. excess water
- C. blood
- D. A and B only

1.1.2. The cells of which structures in the kidney contain relatively large numbers of mitochondria?

- A. The glomerulus
- B. The afferent arteriole
- C. The proximal convoluted tubule
- D. The Bowman's capsule

1.1.3. Which of the following is least likely to be present in the glomerular filtrate (the filtrate produced by the nephron before it enters the loop of Henle) of a healthy adult nephron?

- A. Amino acids
- B. Glucose
- C. Electrolytes
- D. Proteins

1.1.4. Presence of which of the following in the urine, even in small amounts is indicative of a possible problem in the renal system of a patient?

- A. Glucose
- B. Creatinine
- C. Urea
- D. Sodium

1.1.5. The pH of blood plasma varies between 7,35 and 7,45. The pH of urine varies between 4,5 and 8,0. What does this reflect about the work of the kidney?

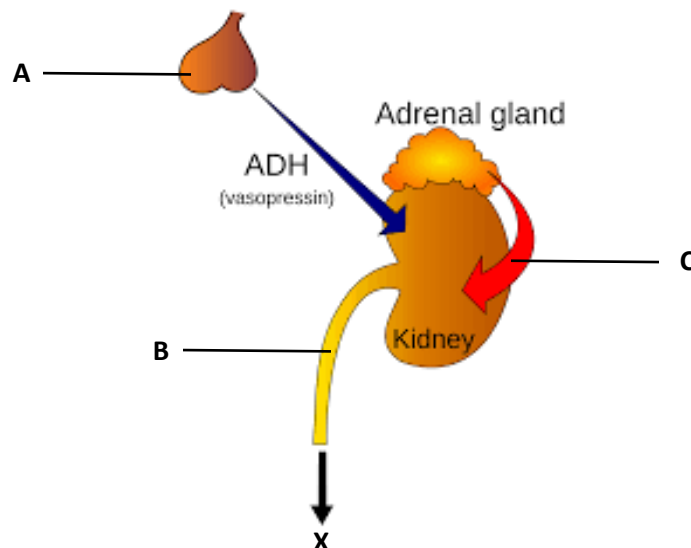
- A. As the pH of urine varies so much, the kidney has very little influence on controlling the pH of blood plasma.
- B. The kidney must play a role in controlling the pH of the blood plasma at a slightly alkaline level.
- C. The kidney is responsible for maintaining the pH of the blood plasma at a slightly acidic level.
- D. The kidney does not monitor the pH levels of the blood plasma.

1.2. Give the correct biological term for each of the following descriptions. Write only the term next to the question number. (5)

- 1.2.1. Concave part of the kidney where blood vessels enter and leave the kidney.
- 1.2.2. Elastic, muscular sac in the pelvic cavity that collects urine prior to excretion.
- 1.2.3. Maintenance of a constant internal environment.
- 1.2.4. The functional and structural unit of the kidney.
- 1.2.5. Specialized epithelial cells that line the Bowman's capsule and facilitate filtration.

Question 2:

2.1. Study the diagram below and answer the questions that follow.



2.1.1. Provide the following:

- A. The names of **structures** A and B. (2)

- B. The **hormone** represented by the arrow at C. (1)
C. The **fluid** represented at X. (1)

2.1.2. ADH has a key function in the homeostasis of water content, known as osmoregulation. Define osmoregulation. (1)

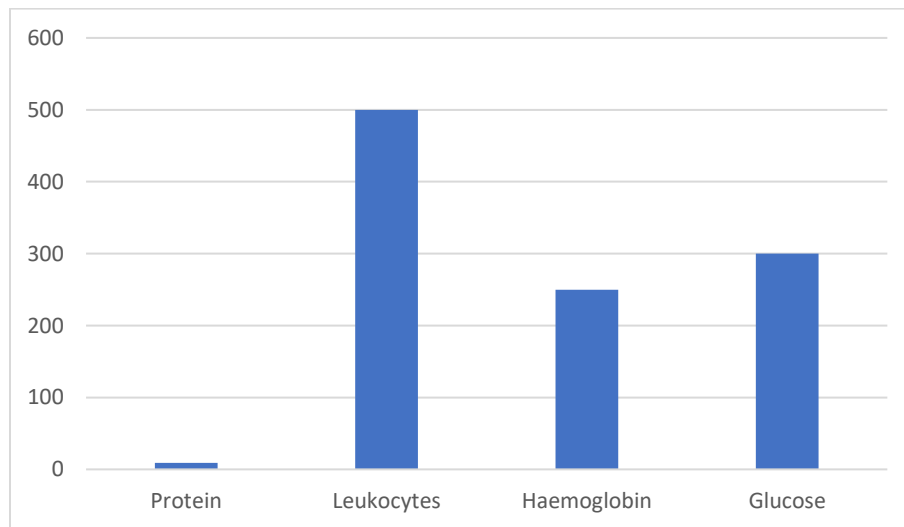
2.1.3. Name the specialized cells that detect and respond to changes in water levels of the blood. (1)

2.1.4. Give the location of the cells mentioned in QUESTION 2.1.3. (1)

2.1.5. Stimulation of the cells in QUESTION 2.1.3. may lead to the secretion of ADH. Briefly explain the role of ADH. (3)

(10)

2.2. A female patient suspects that she may have Cystitis (a UTI that affects a particular part of the urinary system). She goes to her doctor who administers a urine test (Combur-10 Strip Test). The doctor would look for the presence of white blood cells and other symptoms to confirm an infection. The patient's results are presented on the graph below. Study her results and answer the questions that follow.

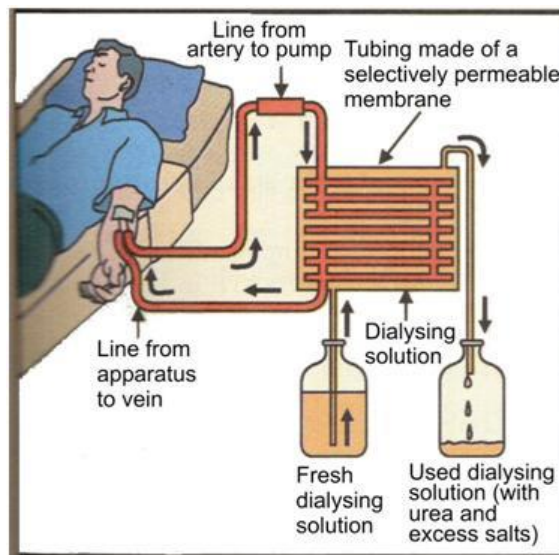


Normal ranges	
Protein	30 mg/dL or less
Leukocytes (white blood cells)	10 or less Leu/ μ L
Haemoglobin	0 Ery/ μ L
Glucose	10 or less mg/dL

- 2.2.1. Which component of the test would be used to confirm the presence of an infection? Explain your answer. (2)
- 2.2.2. Which result was within normal range? (1)
- 2.2.3. Consider the patient's test result for Glucose.
- A. Calculate the difference in concentration between the result and normal range for glucose. (2)
- B. What could the glucose result of the test possibly indicate? (1)
- C. Where in the nephron would the glucose of a healthy female be completely reabsorbed? (1)
- (7)**

Question 3

If a person's kidneys happen to get damaged or do not function correctly a person can be placed on a dialysis machine to help the functioning of the kidneys.

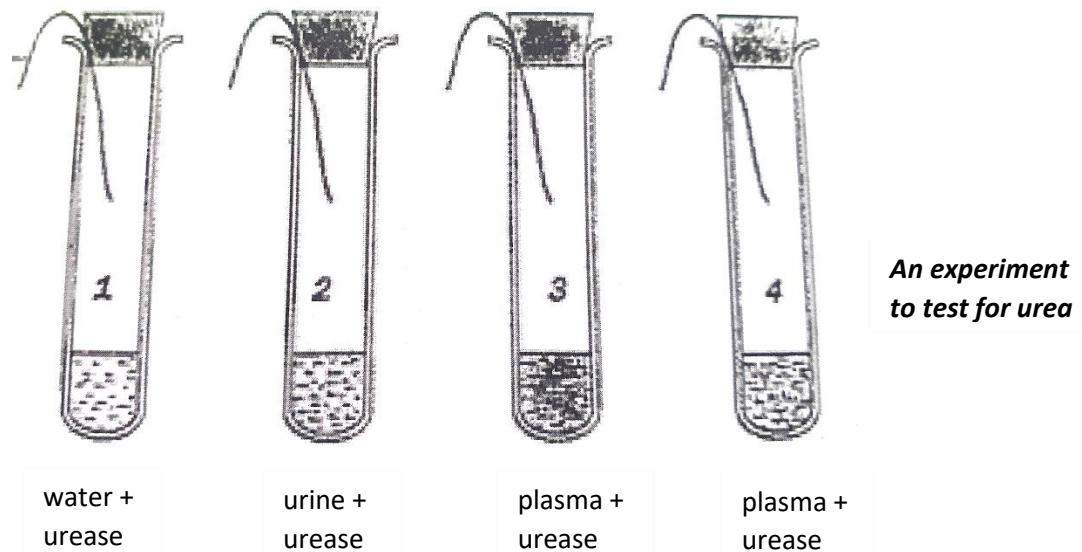


- 3.1. In which part of the nephron does a similar process take place as illustrated by the kidney machine? (1)
- 3.2. Explain the purpose of having so many tubes carrying blood through the dialyser. (2)
- 3.3. By referring to the composition (make up) of the dialysis fluid, explain how the patient's blood is filtered. (4)

- 3.4. Why does dialysis fluid continually need to be replaced? (2)
- 3.5. Provide TWO disadvantages of dialysis. (2)
- 3.6. What is the long-term solution for kidney failure? (1)
- (12)**

Question 4

Read the information provided for the following experiment and answer the questions that follow.



- A. Four test tubes were placed in a test tube rack and labelled 1-4.
- B. A mixture of water and urease was placed in test tube 1.
- C. A mixture of urine and urease was placed in test tube 2.
- D. A mixture of plasma from the blood in the *renal artery* and urease was placed in test tube 3.
- E. A mixture of plasma from the blood in the *renal vein* and urease was placed in test tube 4.
- F. The four test tubes were placed in a water bath at 37°C for 30 minutes.
- G. The mixture in each test tube was tested with a test strip to detect the presence of ammonia. (**urea is broken down into ammonia by the enzyme urease.**)
- H. The results of the experiment are shown in the following table:

Test tube	Test for ammonia
1	Negative
2	Positive
3	Positive
4	Negative

- 4.1. Why do you think the test tubes were kept at 37°C? (2)
- 4.2. Name the independent and dependent variables in the investigation. (2)
- 4.3. Which test tube acted as the control? Give a reason for your answer. (2)
- 4.4. Use the results of this experiment and your own knowledge of urinary system to explain what happens to urea in blood as it travels through the kidneys. (4)
- 4.5. Name ONE variable in the experiment that was controlled? (1)
- (11)**

TOTAL: 50 MARKS